InfraSight

The AI co-pilot for city streets.

A citizen photographs a broken street. Three Meta models turn that one photo into a case the town hall can act on: classified, outlined, priced from sourced rates, routed around, and drafted into legal documents.

- Muse Spark
- SAM 3.1
- Muse Image



A real citizen report: a smashed bus shelter

Meta Global AI Developer Hackathon · November 2026 · working product · github.com/abdo0620/InfraSight

THE PROBLEM

Every broken street is a small administrative maze.

A town hall receives hundreds of reports a year. Each one has to be understood, checked, routed, approved, bought, signed and closed. Meanwhile, citizens hear nothing.

01

Unclear reports

Blurry photos, vague addresses, the same hole reported five times.

02

Who is responsible?

Commune, department, State, intercommunality, network operator or private owner.

03

Rules at every step

Purchase thresholds, council votes, works declarations, traffic orders, acceptance.

04

Silence for citizens

People rarely hear back, so they report again, or stop reporting.

THE IDEA

One photo in. A decision-ready case out.



1 · The citizen's photo
pinned on the map, one sentence



2 · The damage, outlined
SAM 3.1, after Muse Spark's triage



3 · The street repaired
Muse Image, labelled as an illustration

4 · The case, ready

- ✓ Type, severity, owner
- ✓ Report with cited rules
- ✓ Cost priced from sources
- ✓ Detour + traffic order
- ✓ Drafts for 7 legal stages
- ✓ Updates to the citizen

USE OF META AI · CITIZEN REPORT

Three Meta models, live, in front of the citizen.

After "Send report", the citizen watches each model work, step by step, with its result and real timing.

Muse Spark · triage

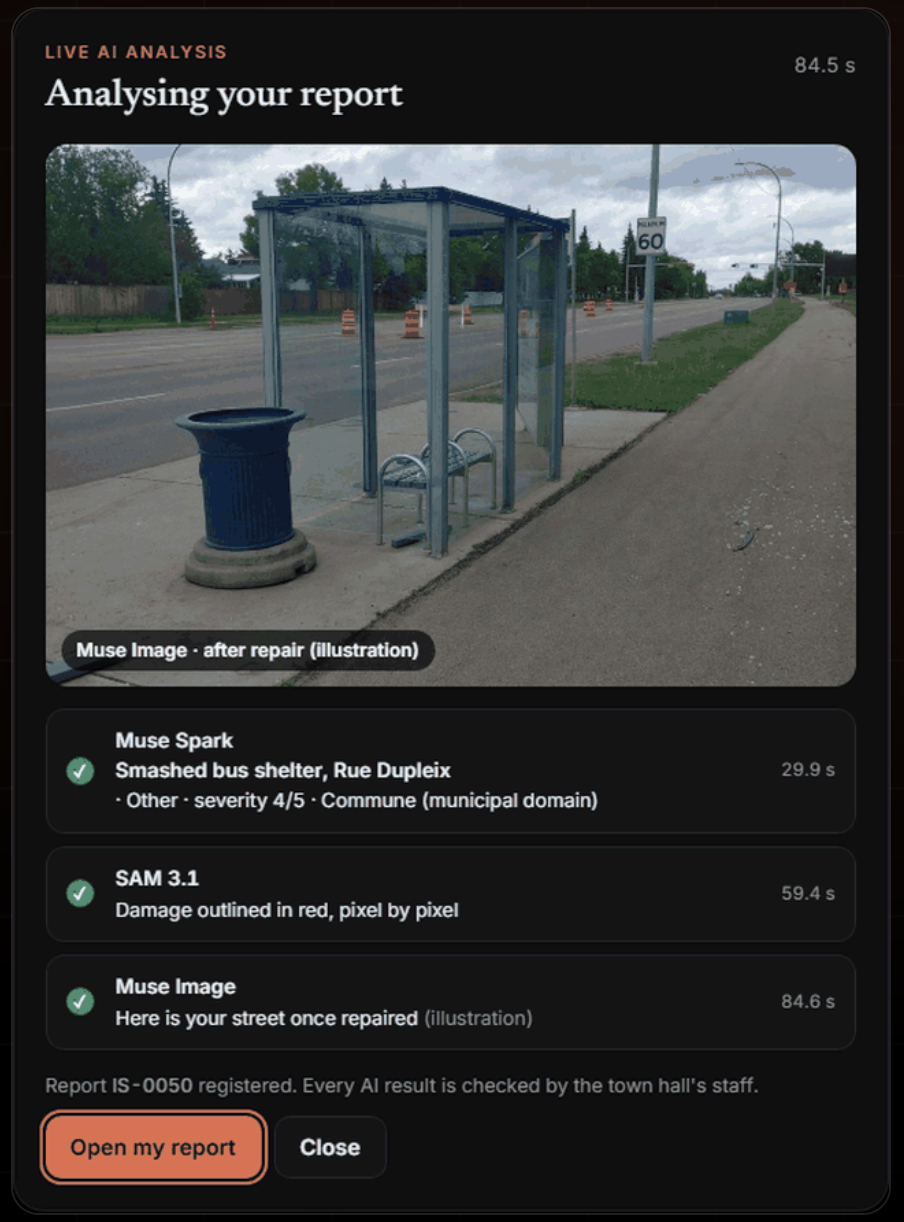
Reads the photo and the words: what it is, severity, urgency, owner, danger zone, duplicates nearby.

SAM 3.1 · outline

Finds the damage by name and outlines it, pixel by pixel.

Muse Image · repair preview

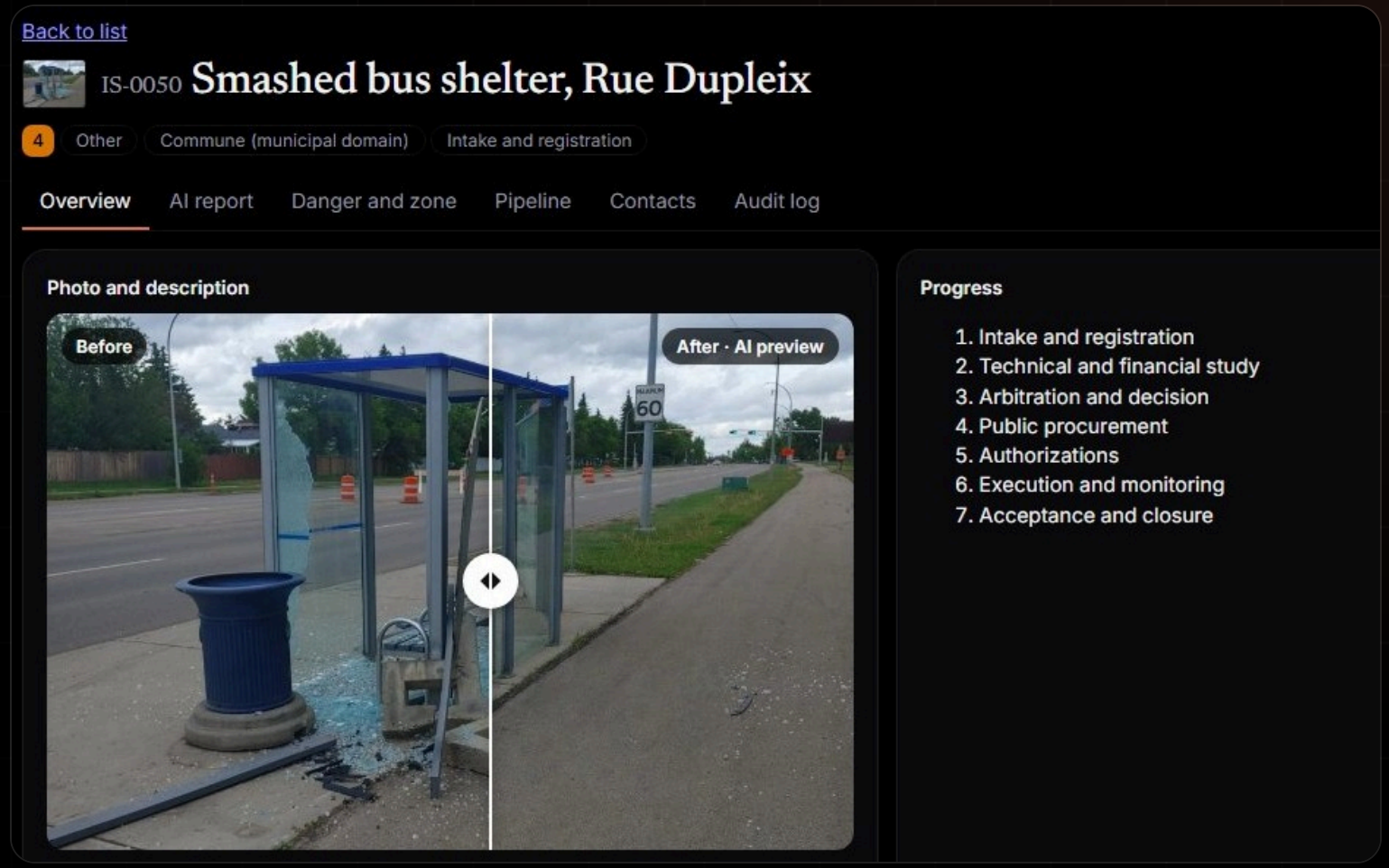
Draws the street once repaired; the real photo is kept outside the damaged area.



Screenshot of the working app · about 85 s end to end · 4

SCREENSHOT OF THE WORKING APP

See the street repaired before work starts.



Before / after slider

Drag to compare the citizen's photo with Muse Image's repair. Once the work is done, the real after-repair photo replaces the illustration.

The same images, everywhere

Ticket, AI report, map popups, the acceptance stage and the PDF report all reuse the photo, the outline and the preview.

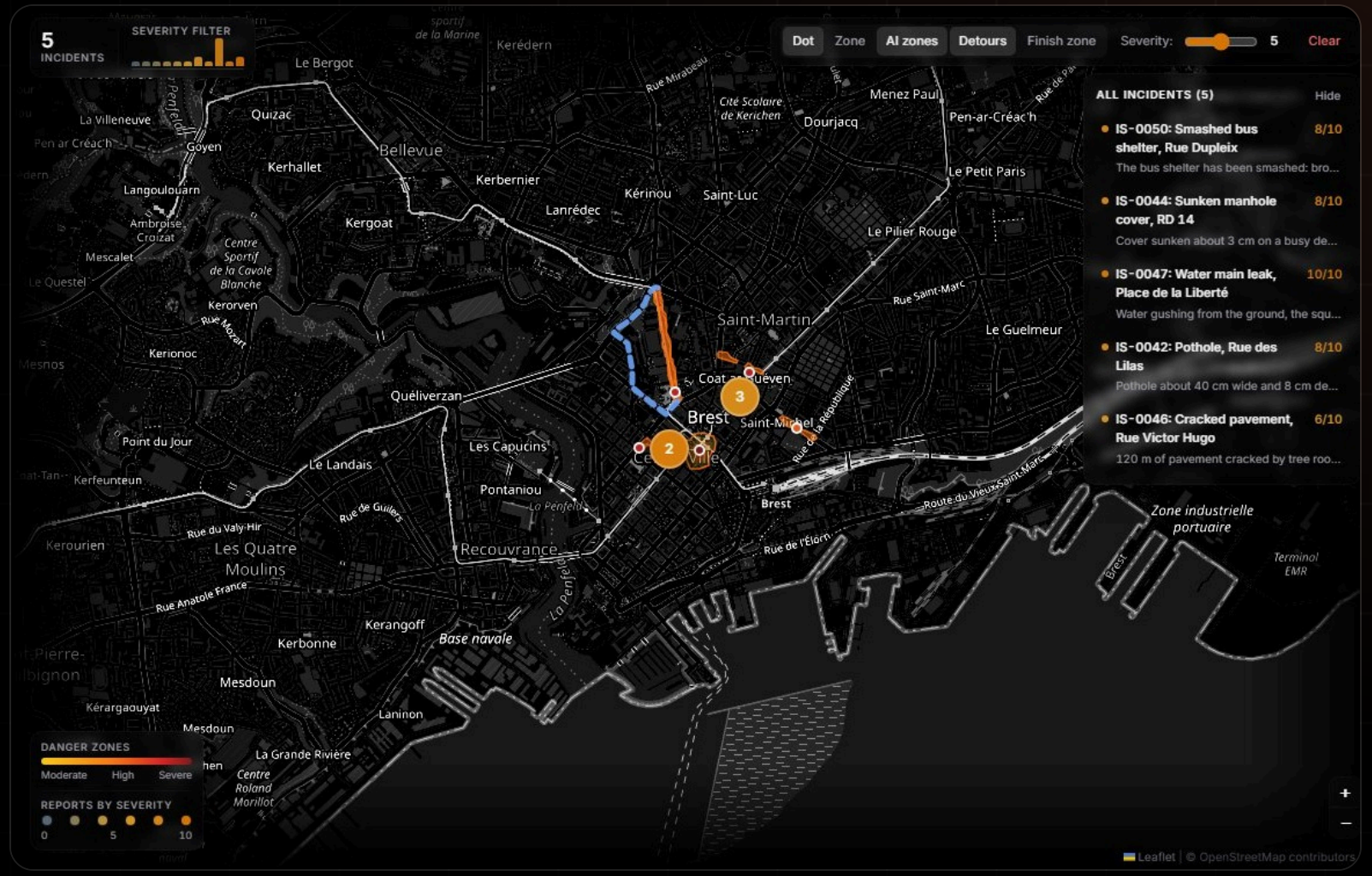
Honest by design

The repair preview is labelled "illustration, not a photo" wherever it appears.

MUSE SPARK · OPENSTREETMAP · STAFF MAP

SCREENSHOT OF THE WORKING APP

Every danger, on one map.



Danger zones

Muse Spark decides the shape: a point, a whole street section (shape from OpenStreetMap) or an area. Yellow to red with the danger.

Detours and hotspots

Blue detours, purple rings on streets where problems keep coming back.

Every report is on the map

A citizen pins the place or picks an address from OpenStreetMap; free-text locations are refused.

MUSE SPARK · DUPLICATE MERGING

Five reports. One case.

When a report lands within 80 m of an open case, Muse Spark is given the nearby cases and decides if it is the same problem. If so, the report joins the case: its photo goes into the history and the citizen follows the case.



Report 1 · 08:12 "Huge pothole by the school"



Report 2 · 09:40 "A cyclist nearly fell"



IS-0042 · Pothole, Rue des Lilas

2 citizens follow this case

same problem, 12 m apart

Less work

One case to process instead of many.

Severity can rise

More reports and photos make a stronger case.

Everyone is told

All citizens who reported it get the same updates.

MUSE SPARK · FULL REPORT · TECHNICAL DEPTH

SCREENSHOT OF THE WORKING APP

A decision-ready report. Every euro sourced.

Cost estimate

Estimate: €844–€890 excl. VAT

Work item	Qty	Rate	Total
Temporary cold-mix patch, to make safe Temporary make-safe of about 1 m2 broken area estimated from photo, ticket notes 40 cm wide core plus surrounding breakup; no damage_measured provided Source: Prix de l'enrobé au m², Habitatpresto (market prices); BPU annexe 2, commune de Prunelli-di-Fiumorbu, 2021	1 m²	€26–€47	€26–€47
Planing (rabotage) of the old asphalt, 1 to 6 cm, under 50 m² Planing required because surface broken over more than 3 cm, ticket states 8 cm deep; same about 1 m2 estimated from photo Source: Grille des réfections de tranchées, Bourg-en-Bresse	1 m²	€16	€16
Final hot-mix asphalt repair, under 10 m² (4 to 8 cm) Final hot-mix repair of same about 1 m2 estimated from photo, under 10 m2 Source: BPU travaux, Régie des eaux Terre de Provence, 2022; BPU annexe 2, commune de Prunelli-di-Fiumorbu, 2021	1 m²	€79–€104	€79–€104
Temporary road signage: set up and removal Set up and removal for works on carriageway and cycle lane near school Source: BPU travaux, Régie des eaux Terre de Provence, 2022	1 lump sum	€578	€578
Temporary road signage: upkeep Upkeep estimated for 5 days until final repair and acceptance Source: BPU travaux, Régie des eaux Terre de Provence, 2022	5 day	€29	€145

No measured size: quantities estimated from the photo [verify]. Run "Outline and measure the damage" for measured quantities. Unit rates from public French price schedules (Prix de l'enrobé au m², Habitatpresto (market prices); BPU annexe 2, commune de Prunelli-di-Fiumorbu, 2021; Grille des réfections de tranchées, Bourg-en-Bresse; BPU travaux, Régie des eaux Terre de Provence, 2022): replace them with the commune's own contract rates.

Recommended actions

Action	Owner	Deadline
Install cones, barriers and warning signs around pothole	Commune – Roads department	within 24 h
Issue temporary traffic and parking order and post on site	Mayor's office	before works

Muse Spark picks the work

Work items and quantities, from the measured damage size. It never writes an amount.

The server prices it

Unit rates from public price schedules (French BPU for now), a source link on every line. A commune plugs in its own contract rates.

Rules cited

Procurement route, council vote, authorizations: each with its article, or marked [verify].

OPENSTREETMAP · MUSE SPARK · REAL STREETS

SCREENSHOT OF THE WORKING APP

The road is closed. Here is the way around.



Consequences if not fixed

Danger: Collision risk, traffic disrupted on RD 14

- Cars swerve into the oncoming lane medium

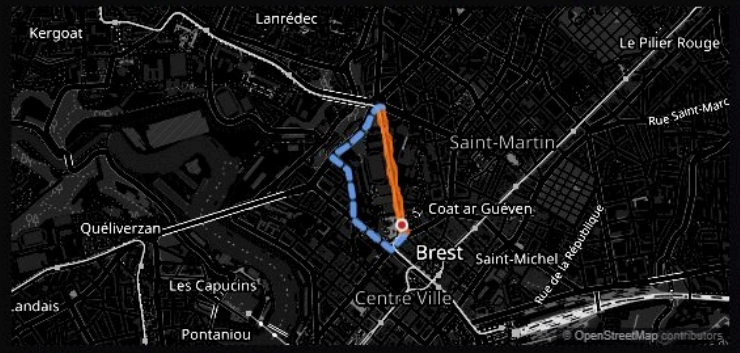
Collision

People at risk: Drivers, cyclists

Immediate measures

- Signage and speed limit
- Raise the cover

Approximate danger zone · along Avenue du Maréchal Foch, high danger



RD 14 section around the cover

Recalculate the detour

Detour (blue dashes): 1021 m via Rue Duquesne → Avenue Georges Clemenceau → Rue de l'Harteloire → Rue Bertrand du Guesclin → Rue Lamotte-Picquet → Place Albert 1er → Avenue Camille Desmoulins

Traffic order, AI draft to check before the mayor signs

[to fill] marks what only the town hall can decide: the order number, the dates and hours of the closure, the signing date. The AI leaves them blank on purpose instead of inventing them. [to check] marks a point to check.

MUNICIPAL ORDER No. [to fill] on temporary traffic regulation - Avenue du Maréchal Foch

The Mayor of Brest,

Having regard to the General Code of Local Authorities (CGCT), in particular articles L2213-1 and L2213-2;
Having regard to the Highway Code (Code de la route);

A real detour

InfraSight downloads the road network around the closed section and computes the shortest legal detour, one-way streets included: here 1,021 m via 7 streets.

The mayor's order, drafted

Muse Spark drafts the temporary traffic order with the detour in it. What only the town hall decides stays [to fill].

Hours of work saved

Staff check it; the mayor signs.

MUSE SPARK · FOR STAFF

The paperwork, drafted. The repair, checked.

7 legal stages, an AI draft for each

- ✓ Intake and registration
- ✓ Technical and financial study
- › **Arbitration and decision** · signed only by an elected official
- ○ Public procurement
- ○ Authorizations
- ○ Execution and monitoring
- ○ Acceptance and closure

Decision memo, specifications, permits checklist, acceptance report: drafted from the case, with the rules and deadlines of each stage.



Before · damage outlined



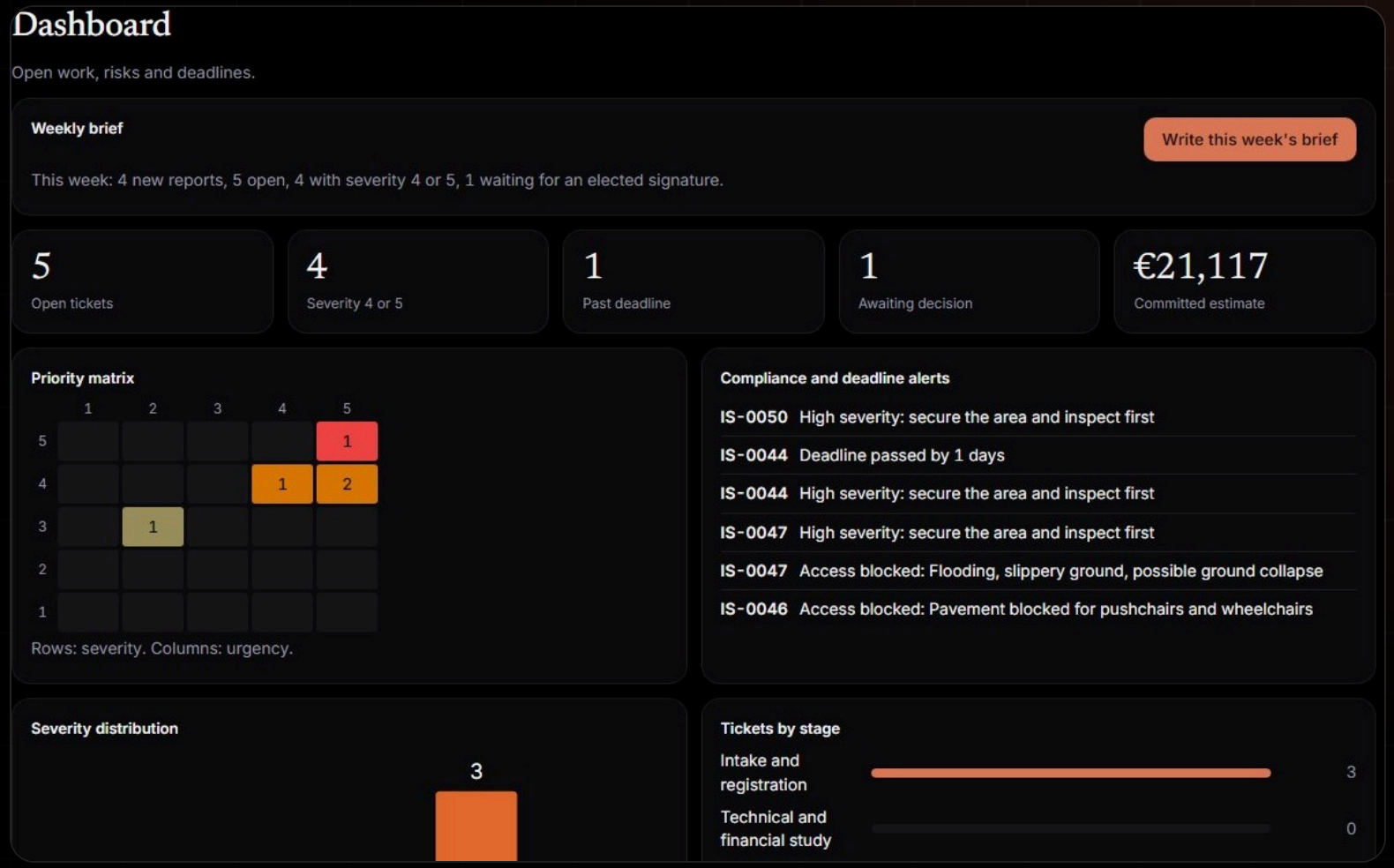
After-repair photo (example)

Verdict: repaired
At acceptance, Muse Spark compares the after-repair photo with the outlined "before". Staff confirm on site.

MUSE SPARK · FOR ELECTED OFFICIALS

SCREENSHOT OF THE WORKING APP

See the whole city. Act before it breaks.



Priorities at a glance

Open cases, severity, deadlines, committed budget and a priority matrix.

A weekly brief

Written by Muse Spark from the week's figures, for the elected officials.

Hotspots and forecast

Streets where problems return; Muse Spark proposes preventive work and a budget.

MUSE SPARK · CITIZENS AND RECORDS

Citizens hear back. The case is on paper.

Updates in the citizen's language

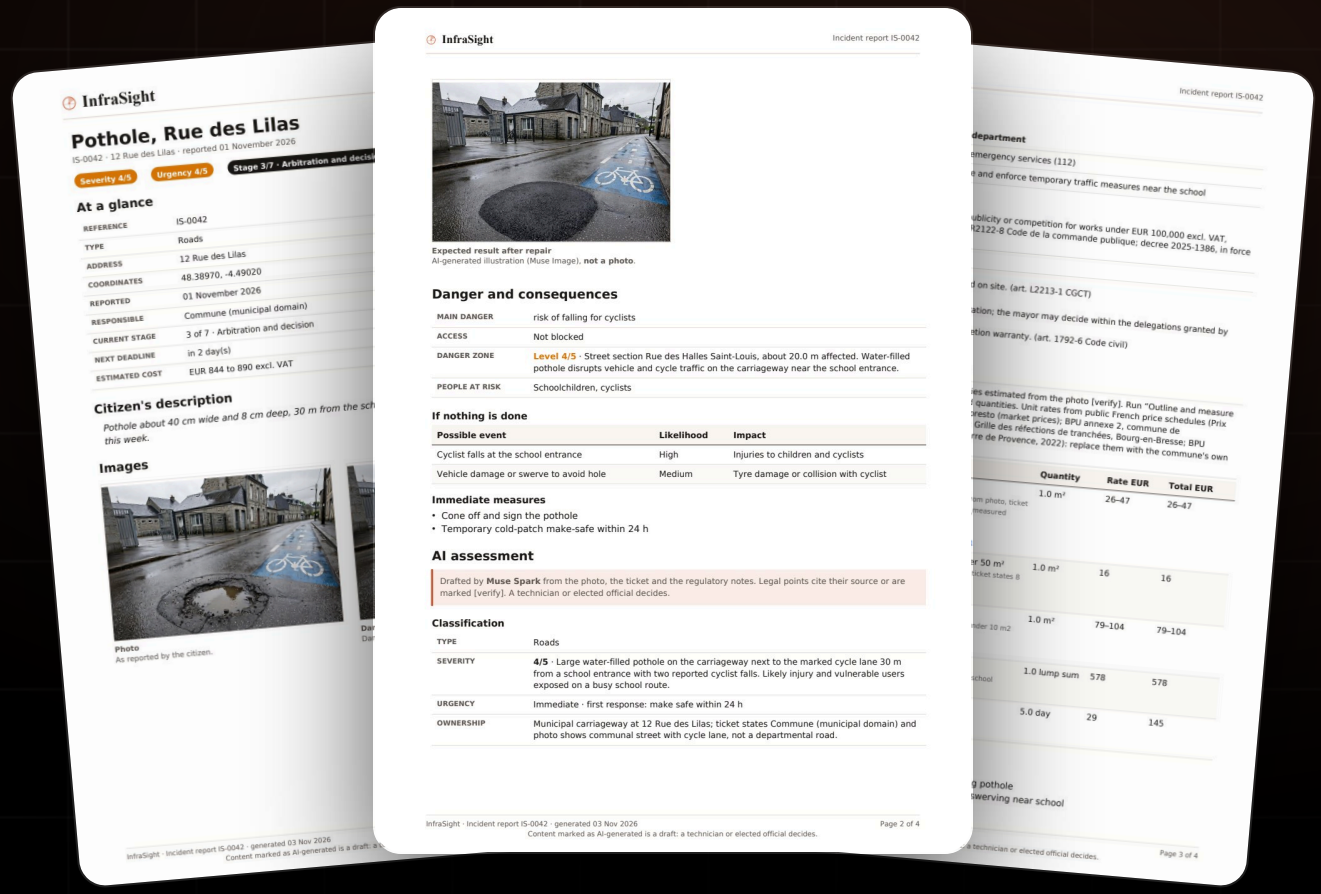
At each step, Muse Spark writes a short, plain update in the language the citizen chose: 9 languages.

"Your report about the bus shelter was received. The roads team will make it safe today."

"Votre signalement de l'abribus a bien été reçu. L'équipe voirie sécurise la zone aujourd'hui."

"Hemos recibido tu aviso sobre la marquesina. El equipo municipal la asegura hoy."

English · Français · Brezhoneg · Español · Português · العربية · Türkçe · Deutsch · Italiano



One-click PDF incident report · images, sources, cost, stages, detour, traffic order

META MODEL API · BEYOND A BASIC INTEGRATION

Powered by Meta. Three models, chained.



One photo + one sentence

from a citizen, pinned on the map

Muse Spark

Triage, duplicates, full report with cited rules, cost work items, damage size, legal drafts, traffic order, repair check, weekly brief, forecast, updates in 9 languages

names the damage

SAM 3.1

Outlines the damage on the photo, from the name Muse Spark gives it, with the triage box as a fallback prompt

outline

Muse Image

Draws the street repaired from the photo and the outline

A decision-ready case

- ✓ Report with sources
- ✓ Sourced cost
- ✓ Detour + order
- ✓ 7 stage drafts
- ✓ Citizen updates
- ✓ PDF report

All on the Meta Model API with one key, no GPU · multimodal from the first step · about $0.02 per citizen report

TECHNICAL DEPTH · TRUSTWORTHY AI

Engineered to be checked, not just prompted.

01

Prompts stay on the server

The browser never sends a prompt. Citizen text is wrapped as untrusted data, so a report cannot steer the model.

02

Validated answers

JSON checked against schemas; an invalid answer is retried once with the exact error.

03

Sources or [verify]

Every legal statement cites its article, or is marked to verify. Amounts come from sourced rates, never from the model.

04

People decide

Severity, money, closures and signatures belong to staff with the right role. Every action goes in the audit log.

05

Live, visible progress

Every AI action shows the real steps the server takes, with results and timings. A failure says which step and why.

06

Resilient

If the AI is down, the report is still saved. Busy map servers are retried, and road data is cached.

FOR THE JUDGES

Built for the four criteria.

Impact — 30%

Every commune handles road, lighting, waste and water reports, and citizens rarely hear back. InfraSight routes each case, drafts the paperwork of each legal stage and keeps citizens informed. Ready for a pilot.

Use of Meta AI tooling — 25%

3 Meta models, 10 AI functions, chained: Muse Spark names the damage for SAM 3.1, the outline guides Muse Image, Muse Spark checks the repair at acceptance.

Demo — 25%

A working product: one command to run (start.bat / start.sh), demo accounts, live AI panels, CI on every push.

Technical depth — 20%

FastAPI with roles and audit log, schema-validated AI, cited sources, OpenStreetMap routing, sourced pricing, PDF reports.

WHERE WE ARE

A working product, ready for a pilot.

Built during the hackathon

- ✓ Citizen, technician, elected roles
- ✓ Muse Spark: 10 AI functions
- ✓ SAM 3.1 outline, Muse Image preview
- ✓ Sourced costs, detours, PDF
- ✓ Danger zones, duplicates, hotspots

Next: a 3-month pilot

One commune, real reports from one neighbourhood · a legal review of every rule · the commune's own contract rates · response times and citizen satisfaction measured

Let's fix streets faster, together.

`github.com/abdo0620/InfraSight`



The same shelter, repaired by Muse Image (illustration)